AMMAR YUSRI

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EDUCATION

Singapore University of Technology and Design Sep. 2022 – May. 2026

- *Bachelor of Science (Design and Artificial Intelligence), Computer Science Minor*

EXPERIENCE

Applications & IT Intern Jan. 2024 – Sep. 2024

LBBW (Landesbank Baden-Württemberg)

- Developed 6 Power BI dashboards to analyse asset lifecycle, warranty status, and compliance, improving decision visibility across departments
- Automated data pipelines using APIs (FreshService, Intune, Power BI), reducing manual reporting effort by 30%
- Applied data modelling and ETL concepts through Microsoft Fabric and Azure AI training, strengthening foundation in scalable data systems
- Enabled real-time reporting workflows, improving operational efficiency and reducing latency in asset tracking
- Designed and executed a 4-tier evaluation framework (15 test cases) to reduce hallucinations and validate model performance

PROJECTS

Infineon Technical Assistant Chatbot (2026) | *AI / RAG System*

Partnered with Infineon to develop a technical assistant for developers using their products

- Architected a dual-model Retrieval-Augmented Generation (RAG) system using OpenAI + BGE embeddings over 88,000+ vectors in PostgreSQL (pgvector)
- Built an end-to-end data pipeline ingesting 66,000+ forum posts and 2,000+ GitHub repositories, enabling scalable knowledge retrieval
- Engineered a GPT-4 orchestration layer with NER-based ranking and fallback mechanisms, improving response accuracy and code reliability

URA Prompt to 3D Massing Generation Tool (2025) | *Generative AI*

Partnered with URA to develop a web app that generates 3D urban massing on a defined basemap

- Designed an AI-driven pipeline that converts text prompts into structured urban layouts, accelerating design iteration speed up to 30×
- Implemented multi-stage generation pipeline (road network → parcel generation → 3D modelling), improving automation of urban planning workflows
- Processed and integrated Singapore geospatial datasets, enhancing model realism and alignment with real-world constraints
- Applied Generative AI techniques to produce context-aware outputs for planning and design decision support

NGS Strength Through Art AI-powered Web Application (2024) | *GenAI + NLP*

Partnered with National Gallery of Singapore to develop a digital version of their StArt Programme

- Integrated Google Gemini (GenAI) to analyse user-generated text and generate personalised reflective prompts
- Deployed application on Google Cloud Platform, implementing containerisation and scalable backend infrastructure
- Built full-stack system (React + Rails) supporting AI-driven user interaction and feedback loops

Sony's Angels (2024) | *AI + Data Analytics*

Identified opportunities for design improvements using data-driven text analytics and image processing

- Applied NLP and data scraping pipelines to extract insights from reviews and social media data
- Used RAG-based techniques to identify product improvement opportunities, linking user sentiment to actionable design decisions
- Combined text analytics and generative AI to support data-driven product strategy

CO-CURRICULAR ACTIVITIES

SUTD Muslim Society Feb. 2022 – Feb. 2025

Publicity Director & Logistics Director

- Designed posters, created 60% more digital content, and managed social media accounts, while reducing club expenditures by 20% awareness and engagement
- Facilitated seamless communication with Mendaki, Association of Muslim Professional (AMP), Al-Falah Mosque and other university Muslim societies

SKILLS

AI / Machine Learning: NLP, Retrieval Augmented & Generation (RAG), Embeddings, LLM Orchestration
Data Engineering: SQL, Python, PostgreSQL, ETL pipelines, Data Modeling
Tools & Frameworks: OpenAI API, Google Gemini, Power BI, Microsoft Fabric, Azure AI
Software & Systems: React, Ruby on Rails, Docker (GCP)